

Annual Puget Sound Recreational Crab Creel Report: Summer 2024



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Table of Contents

Acknowledgements	ii
Table of Contents.....	iii
List of Tables.....	v
List of Figures	vii
List of Appendices.....	viii
Executive Summary	1
Introduction.....	2
Dungeness crab fisheries in Puget Sound.....	2
Puget Sound marine areas and seasons	3
Creel Survey Objectives	5
Methods.....	6
Site selection and sampling days	6
Focus area survey.....	6
General compliance survey	7
Creel interview process.....	11
Mean Catch per Harvester Trip	12
Descriptive statistics and analysis	13
Results.....	14
MAs 8-1 and 8-2 Focus Area Survey	14
Sampling frequency	14
Contact types	14
Temporal variability in vessels contacted	17
Compliance – Catch Record Cards (CRC) and unrecorded crab	19
Compliance – Dungeness crab catch	21
Compliance – red rock crab catch.....	21
Catch and effort patterns	22
Weight measurements by marine area	26
Lost recreational crab gear.....	27
General Compliance Survey in all other MAs	28
Pier sampling	31

Summary	34
Appendix B.....	39

List of Tables

Table 1. Summary of recreational seasons by MA for the summer 2024 Dungeness crab fishery.	5
Table 2. Probability weights by MA 8-1 and 8-2 creel locations used to randomly select sites to survey each open fishery day. The proportion of total contacts represents the proportion of contacted vessels by WDFW in 2017 that uses each site.	9
Table 3. Probability weights used to randomly select the shift type strata after a creel location has been selected for MAs 8-1 and 8-2 by the day of the week. Probability weights for fishery day and shift types were calculated using data from creels conducted by WDFW in 2017.	10
Table 4. Summary of creel sampling frequency across the 17 sites surveyed in MAs 8-1 and 8-2 in 2024. Dungeness sampled and red rock crab sampled refers to the number of crabs sampled for compliance (i.e., checking for female, sub-legal, and soft-shell) and excludes catch from vessels where only a portion of the total catch was sampled. This table excludes data from two non-vessel contacts. AM shifts occurred between 08:00 and 14:00. PM shifts occurred between 14:00 and 20:00.	15
Table 5. Total number of vessels contacted at each site by Marine Area of harvest. For a specific creel location, dark color shade indicates a larger proportion of vessels were observed fishing in that MA. ...	17
Table 6. Total number of vessels contacted, vessels with Dungeness crab in possession, and CRCs checked by creel staff, percent of vessels contacted with no CRCs in possession, and percent of vessels that did not comply with the request to view CRCs.	20
Table 7. The number and percent of Dungeness crabs retained by recreational harvesters that were not recorded on Catch Record Cards (CRCs) prior to the time of the interview conducted by WDFW creel staff. The percent of Dungeness crab recorded on summer CRCs is calculated as the total number of unrecorded Dungeness crab divided by the total number of Dungeness crab retained. Vessels contacted that were in possession of Dungeness crab, but CRCs were not checked for reasons other than non-compliance were removed from calculations of percent unrecorded catch.	21
Table 8. Sample size and percent occurrence of retained female, sub-legal, and soft-shell Dungeness and red rock crab by recreational harvesters in MAs 8-1 and 8-2. The percent of female, sub-legal, and soft-shell crab, indicated in '()', is expressed as the total number of females, sub-legal, or soft-shell crab sampled from recreational harvesters within a MA divided by the total number of Dungeness or red rock crab sampled within a MA. Red rock crabs were not assessed for soft-shell. Catch from vessels where only a portion of the total catch was sampled is excluded.	22
Table 9. Summary of the mean number of harvesters per vessel, gear fished per vessel, Dungeness caught and retained per vessel, Dungeness caught and retained per individual harvester trip, and the proportion and ratio of red rock crab retained to Dungeness crab by MAs 8-1 and 8-2. The standard deviation associated with each mean value is indicated in '()'. A harvester is defined as a crabber on board the contacted vessel that was engaged in crabbing. A unit of gear includes pots, ring nets, and star traps. Red rock crab expressed as the percent of total crab retained is calculated as the total number of red rock crab retained by recreational harvesters divided by the sum of the total number of red rock crab and Dungeness crab retained. The catch ratio is calculated as the total number of red rock crab observed divided by the total number of Dungeness crab observed.	24

Table 10. Sample size of the total number of crabs weighed, total pounds observed, and mean pounds per individual Dungeness and red rock crab by MAs 8-1 and 8-2. The mean weight per crab for each MA was calculated by dividing the total pounds by the total number of individual crabs weighed. The standard error around the mean weight per crab was estimated using a bootstrap procedure ($N = 2,000$). Catch from vessels where only a portion of the total catch was weighed is excluded.	26
Table 11. Summary of recreational crab gear loss, including the total number and percent of vessels contacted that reported at least one unit of gear lost during their trip, total units of gear reported lost, total units of gear that had been fished across all contacted vessels in each MA, and the percent of gear fished that was reported lost. A unit of crab gear is defined as either a crab pot, ring net, or star trap. .	27
Table 12. Summary of creel effort (creel hours), number of vessels contacted, number of CRCs checked, and number of Dungeness and red rock crab sampled for compliance and biological data by creel locations sampled opportunistically during general compliance creels.....	29
Table 13. Total number of vessels contacted at each site by Marine Area of harvest. For a specific creel location, dark color shade indicates a larger proportion of vessels fished in that MA. Only creel locations where at least 10 vessels were contacted are included.....	30
Table 14. Sampling frequency of recreational harvesters at pier locations, number of harvesters (or harvester groups) interviewed, and number of Catch Record Cards (CRCs) checked during the summer 2024 Puget Sound recreational crab season. Only harvesters with Dungeness crab in possession were asked to view their CRC.	31
Table 15. The number of Dungeness crabs retained, recorded, and unrecorded on Catch Record Cards (CRCs) prior to the time of the interview by recreational harvesters at piers.....	32
Table 16. Sample size and percent occurrence of retained female, sub-legal, and soft-shell Dungeness and red rock crab by recreational harvesters at each pier location. The percent of female, sub-legal, and soft-shell crab, indicated in '()', is expressed as the total number of females, sub-legal, or soft-shell crab sampled from recreational harvesters at a pier location divided by the total number of Dungeness or red rock crab sampled.	33

List of Figures

Figure 1. Map of each MA found within Puget Sound, Washington. Puget Sound MAs include areas 4 (east of the Bonilla-Tatoosh line), 5, 6, 7, 8-1, 8-2, 9, 10, 11, 12, and 13.	4
Figure 2. Map of creel sampling locations (red points) in Puget Sound’s MAs 8-1 and 8-2.	8
Figure 3. The number of 6-hour creels conducted throughout the 2024 summer crab season (date; x-axis figure a) and percent of the total crabbing vessels contacted (x-axis figure b) at each creel site (y-axis). In figure a, each point corresponds to an individual 6-hour creel conducted by WDFW sampling staff. Color represents the day of the week the creel was conducted, and the shape of each point corresponds to the shift type selected. In figure b, each bar represents the percent of the total number of vessels contacted at each creel location.	16
Figure 4. Mean number of vessels contacted at the four most frequently sampled sites by hour bin (2400-hour clock) for the summer 2024 recreational crabbing season.	18
Figure 5. Mean vessels contacted at the four most frequently sampled sites by day of the week and shift strata. Numbers on top of each bar represent the number of creels conducted for a unique weekday and shift strata.	19
Figure 6. Interval plots showing mean crabbers per vessel, mean units of gear fished per vessel, mean number of Dungeness caught and retained per vessel, and mean number of Dungeness caught and retained per individual harvester trip by MA (x-axis). A harvester is defined as a crabber on board the interviewed vessel that was engaged in crabbing. A unit of gear includes pots, ring nets, and star traps. Red points indicate the mean, gray points indicate observed values, and vertical black lines above and below the mean represent $\pm$ standard deviation.	23
Figure 7. Dungeness and red rock crab expressed as percent of total crab retained by recreational harvesters by MA.	25

List of Appendices

Appendix A. 2024 Puget Sound Recreational Crab Creel Sampling protocol.....	35
Appendix B. Sampling instructions for checking for soft-shell Dungeness crab.	39

Executive Summary

The Washington Department of Fish and Wildlife (WDFW) Puget Sound Shellfish Team conducts creels of recreational crab harvesters in Puget Sound to monitor fishery compliance, effort and catch rates, and promote education within the Puget Sound recreational Dungeness crab (*Metacarcinus magister*) and red rock crab (*Cancer productus*) fisheries. Along with monitoring baseline fisheries information, such as harvest rates, fishery compliance was documented and evaluated. This includes collecting data to better understand the extent of catch unrecorded on a Puget Sound Catch Record Card, and documenting the retention of female, sub-legal (below the minimum carapace width), and soft-shell crab Dungeness and red rock crab to identify compliance issues and educational needs. Boat ramps, marinas, and piers with a range of high to low use volume were monitored each open fishery day during the summer 2024 crab season (July 1 - September 2). In 2024, Puget Sound recreational crab creels followed a stratified random survey design to conduct creels at several sites in Puget Sound's Marine Areas 8-1 and 8-2. In Marine Areas 8-1 and 8-2, the survey was stratified by site (n = 17 sites) and shift type (morning vs. afternoon) and were randomly assigned to WDFW creel staff each open fishery day based on a probability of each stratum being selected. Limited staff availability prevented surveying other Puget Sound Marine Areas following a stratified random sampling design and were instead sampled opportunistically.

Across all Marine Areas surveyed during the summer 2024 crab season in Puget Sound, 318 individual creels were conducted across 38 different sites. A total of 4,710 motorized and non-motorized vessels and an intertidal wader and a group of scuba divers engaged in recreational crabbing in Puget Sound were contacted. Of the 36,011 Dungeness crabs observed in possession, 52% (n = 18,704) were sampled for biological and compliance data and 38% (n = 13,534) were weighed to assess the average weight of an individual Dungeness crab. Of the 4,362 red rock crab retained, 39% (n = 1,719) were sampled for biological and compliance data and 24% (n = 1,056) were weighed to inform the average weight of an individual red rock crab. Additionally, 8,516 Catch Record Cards were checked from individuals that caught and retained Dungeness crab to better understand recording compliance.

This report summarizes findings from the 2024 Marine Area 8-1 and 8-2 random stratified survey, followed by a summary of data collected in other Puget Sound Marine Areas which were sampled opportunistically.

Introduction

Dungeness crab fisheries in Puget Sound

The Dungeness crab (*Metacarcinus magister*) is native to Washington State and supports highly valued state commercial, state recreational, and tribal commercial, ceremonial, and subsistence fisheries. Dungeness crab range from central California to the Gulf of Alaska, with the range of distribution in Washington State encompassing the waters of Puget Sound, Strait of Juan de Fuca, San Juan Archipelago, Hood Canal, and Washington's coastal areas. The Dungeness crab fishery in Puget Sound is managed separately from the Washington coastal fishery, reflected by differences in harvest rules, season lengths, and catch reporting requirements. Further, shellfish in Washington are co-managed by the State of Washington through the Washington Department of Fish and Wildlife (WDFW) and treaty tribes in Puget Sound and coastal Washington under Sub-proceeding 89-3 of *US v. Washington* (Rafeedie decision; 1994). Through these treaties, signatory tribes reserved the rights to harvest up to 50% of the naturally occurring surplus shellfish within their "usual and accustomed" areas.

The Washington Fish and Wildlife Commission's Puget Sound crab policy (C-3609) directs WDFW staff to maintain a healthy Dungeness crab population while providing both recreational and commercial fishing opportunities. The policy defines a base summer recreational season that typically begins in July and extends through Labor Day; 5 days per week, open Thursday through Monday. If quota remains following the base summer season, a winter recreational season may be conducted; typically occurring early October through December 31, 7 days per week. In Puget Sound, the daily recreational limit for Dungeness crab is 5 per day and the minimum size limit is 6 ¼ inches across the width of the carapace. All female and softshell Dungeness crab must be released. In addition, all fishers participating in the State recreational Dungeness crab fishery in Puget Sound must possess a Dungeness crab Catch Record Card (CRC) and immediately record each Dungeness crab retained, including the month, day, and marine area where each crab was harvested. Separate summer and winter CRCs are issued each season of each year and WDFW requires CRCs to be reported each season by the reporting deadline. Fishers have the option to report their catch online during an online reporting period available at the end of each season or mail their CRC back to the Department during or following the season but prior to the reporting deadline. Alternatively, CRCs can be dropped off at a regional Department office prior to the reporting deadline. This information provides valuable data to WDFW and is used to estimate recreational harvest of Dungeness crab in Puget Sound each season.

Puget Sound marine areas and seasons

Recreational fishing opportunities in the marine waters of Washington are spatially managed using Marine Areas (MA). In Puget Sound, there are eleven discrete Marine Areas (MA), each with Dungeness crab seasons that typically vary between the areas (Figure1; Table 1). During the summer 2024 crab season, Area 13 and the portion of Area 12 south of Ayock Point were closed for the entire season as part of continuing annual abundance-related conservation closures. Areas 10 and 11 were open July 1 through August 26 and crabbing was only allowed on Sundays and Mondays. Area 7 South (7S) was open July 18 through September 2 and crabbing was allowed Thursday through Monday. Area 7 North (7N) was open August 15 through September 2 and crabbing was allowed Thursday through Monday. All remaining areas were open July 1 through September 2 and crabbing was allowed Thursday through Monday. In all areas except 10, 11, 12 (South of Ayock Point), and 13, crabbing was allowed on Tuesday, August 27, and Wednesday, August 28.

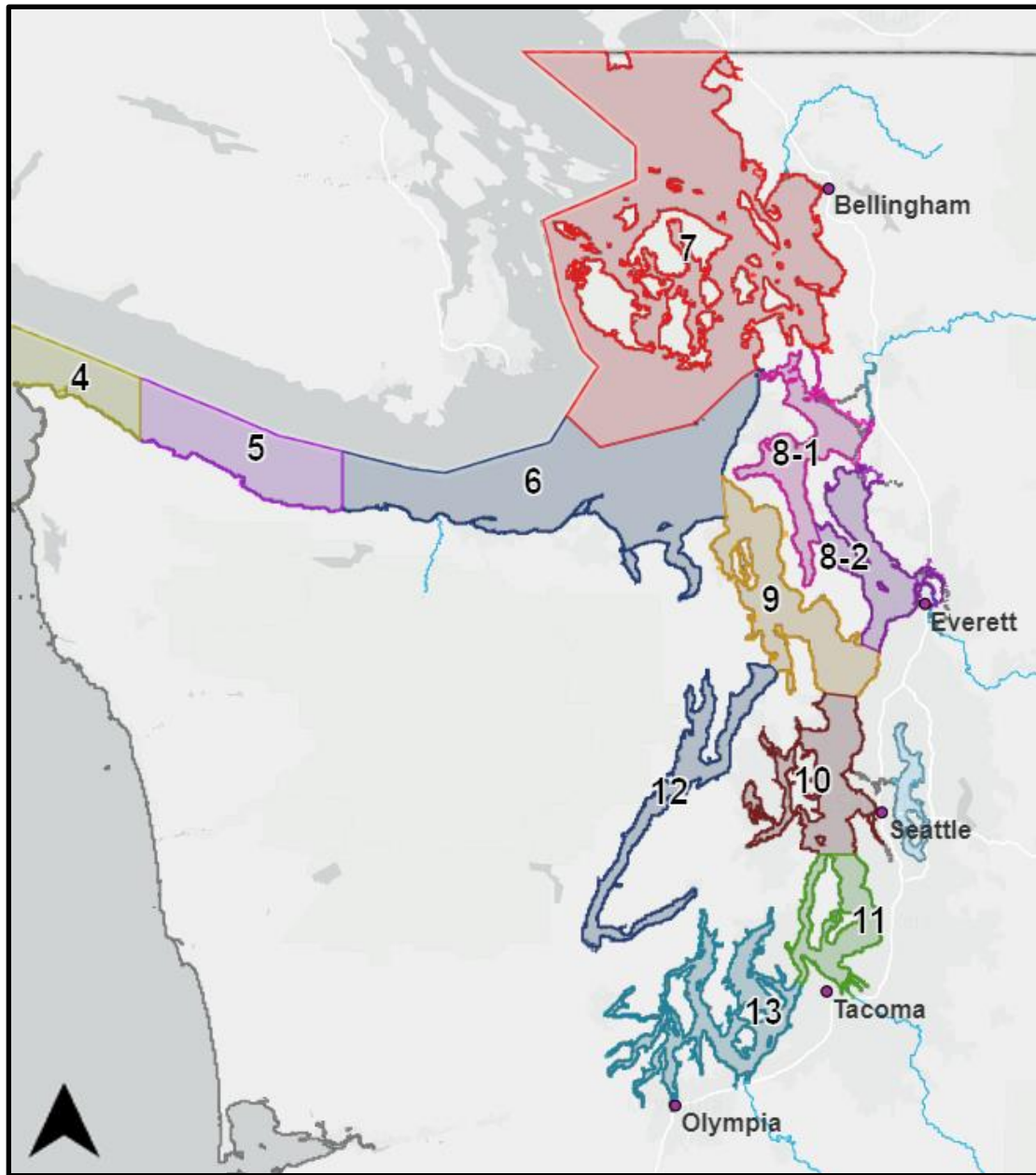


Figure 1. Map of each MA found within Puget Sound, Washington. Puget Sound MAs include areas 4 (east of the Bonilla-Tatoosh line), 5, 6, 7, 8-1, 8-2, 9, 10, 11, 12, and 13.

Table 1. Summary of recreational seasons by MA for the summer 2024 Dungeness crab fishery.

Marine Area	Season	Weekdays open	Total days Open
4	July 1 - September 2	Thursday – Monday**	48
5	July 1 - September 2	Thursday – Monday**	48
6	July 1 - September 2	Thursday – Monday**	48
7 South	July 18 - September 2	Thursday – Monday**	37
7 North	August 15 – September 2	Thursday – Monday**	17
8-1	July 1- September 2	Thursday – Monday**	48
8-2	July 1- September 2	Thursday – Monday**	48
9	July 1- September 2	Thursday – Monday**	48
10	July 1 – August 26	Sunday - Monday	17
11	July 1 – August 26	Sunday - Monday	17
12*	July 1- September 2	Thursday – Monday**	48
13	None	None	0

**The portion of MA 12 south of Ayock Point was closed to recreational crabbing for summer 2024.*

***All areas except 10, 11, 12 (the portion south of Ayock Point), and 13 were open on August 27 (Tuesday) and August 28 (Wednesday).*

Creel Survey Objectives

The annual Puget Sound summer crab creel survey is structured to address key data limited questions in the recreational crab fishery. The focus includes understanding the proportion of unreported catch (i.e., not recorded on CRCs), collecting additional data on harvest and effort, and addressing compliance challenges in Puget Sound’s recreational crab fishery. Due to the spatial complexity and large number of access points to Puget Sound, each annual Puget Sound crab creel may only focus on a few select Marine Areas. In 2024, the annual Puget Sound recreational crab creel focused on Marine Areas 8-1 and 8-2. The objectives are structured as:

1. Collect data on the proportion of recreationally harvested Dungeness crab that are not recorded on Catch Record Cards.
2. Collect data to better understand trends in harvest and catch rates, the timing of effort, amount of gear fished per vessel trip, the average weight per individual crab, and catch compliance in the Puget Sound recreational Dungeness crab fishery.
3. Better understand harvest of red rock crab that has historically been data limited in Puget Sound’s recreational fishery.
4. Continue to establish a presence at access points to promote compliance through education and outreach.

Methods

Site selection and sampling days

Focus area survey

During the 2024 Puget Sound recreational crab season (July 1 – Sept. 2), recreational crab creels followed a weighted random survey design to intercept and interview recreational crab harvesters at several sites in Puget Sound’s Marine Areas 8-1 and 8-2. Limited staff availability prevented surveying other Puget Sound Marine Areas following a weighted random sampling design and were instead sampled opportunistically. In Marine Areas 8-1 and 8-2, the survey was stratified by site ($n = 17$ sites) and shift type (morning vs. afternoon) and were randomly assigned to WDFW creel staff each open fishery day (Thursday – Monday) based on a probability of each stratum being selected. A total of 17 boat ramps and marinas across MAs 8-1 and 8-2 were identified as candidate creel sites (Figure 2). Candidate sites were selected based on data collected in 2017 from a survey conducted by WDFW to better understand where recreational crab harvesters access the water from and the proportion of the crabbing population that uses each access site. The survey found that 73% of the total encountered crabbing effort in MAs 8-1 and 8-2 had accessed the water through 17 sites, while the other 27% of sites represented unknown, private, or logistically challenging sites for WDFW staff to access. Due to staff availability and limitations, 3 sites were randomly selected to survey on Thursdays and Mondays when crabbing effort is assumed to be lowest, whereas 6 sites were randomly selected to survey on Fridays, Saturdays, and Sundays. Due to the wide variation in crabbing effort between sites, site selection was based on a probability weight to account for sampling bias and to ensure the frequency at which each site is selected is representative of the proportion of the population that uses each site (Table 2). Site-specific probability weights were calculated based on the percentage of total on-the-water boat encounters that occurred at each site, as determined by the 2017 survey data. After a site gets selected, the survey is further stratified by shift type (morning vs. afternoon). A typical creel day consists of an 8-to-10-hour workday (including drive and office time), thus only a portion of the day at a selected site can be sampled by an individual creel sampler. Data from past WDFW creels suggest the mean percentage of the total daily crabbing effort occurs between 08:00 and 20:00, thus a morning creel shift occurs from 08:00 to 14:00, whereas an afternoon creel shift occurs from 14:00 to 20:00. Due to differences in crabbing effort between the mornings and afternoons and between weekdays and weekend days, selection of the shift type was weighted by a probability (Table 3). Shift type probability weights were informed by creel data collected by WDFW in 2017. Previous creel data indicates that afternoon shifts had a slightly higher proportion of total boat contacts compared to morning shifts on weekdays (Thursday, Friday, Monday) and a much

higher proportion of total boat contacts in the afternoon compared to mornings on weekend days (Saturday and Sunday).

General compliance survey

Sampling at high volume boat ramps was the priority for opportunistic sampling in MAs to maximize interactions with recreational harvesters. Creel sites were defined as “high volume” using data from past WDFW creels to inform the relative number of users at each site, the overall size and capacity of the site, and the sites proximity to populated areas. Sampling occurred only on days the fishery was open. For MAs 6, 7, 9, and 12 that were open Thursday-Monday, sampling was conducted Thursday-Monday. For MA 10 and MA 11, sampling was focused on both Sundays and Mondays when the fishery was open. Creel sites identified within MA 4 (east of the Bonilla-Tatoosh Line) and MA 5 were not sampled for logistical reasons and MA 13 was not sampled due to continued abundance related closures in this area. Additionally, limited sampling occurred at several piers within Puget Sound to characterize catch and compliance of this user group.

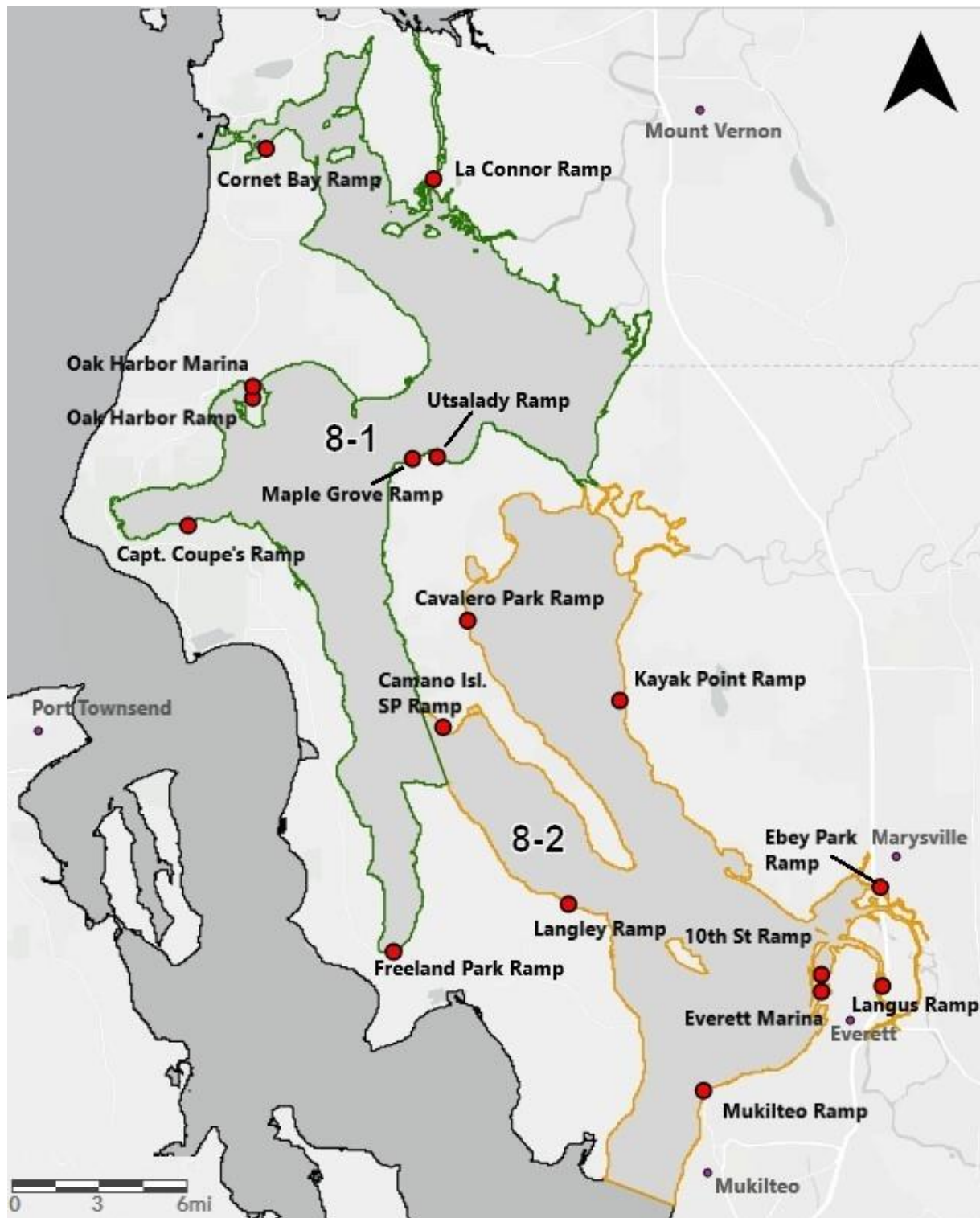


Figure 2. Map of creel sampling locations (red points) in Puget Sound's MAs 8-1 and 8-2.

Table 2. Probability weights by MA 8-1 and 8-2 creel locations used to randomly select sites to survey each open fishery day. The proportion of total contacts represents the proportion of contacted vessels by WDFW in 2017 that uses each site.

Site	Proportion of total contacts	Probability Weight
Everett 10th St Ramp	0.245	0.3365
Cornet Bay Ramp	0.106	0.1456
Camano Island SP Ramp	0.096	0.1319
Kayak Point Ramp	0.044	0.0604
Everett Marina	0.044	0.0604
Oak Harbor Ramp	0.035	0.0481
Maple Grove Ramp	0.031	0.0426
Mukilteo Ramp	0.025	0.0343
Oak Harbor Marina	0.025	0.0343
La Connor Ramp	0.021	0.0288
Ebey Waterfront Ramp	0.019	0.0261
Captain Coupes Ramp	0.010	0.0137
Utsalady Ramp	0.008	0.0110
Cavalero County Park Ramp	0.008	0.0110
Langley Ramp	0.005	0.0069
Langus Riverfront Ramp	0.004	0.0055
Freeland Park Ramp	0.002	0.0027
Total	0.728	1

Table 3. Probability weights used to randomly select the shift type strata after a creel location has been selected for MAs 8-1 and 8-2 by the day of the week. Probability weights for fishery day and shift types were calculated using data from creels conducted by WDFW in 2017.

Fishery Day	Shift Type	Probability Weight
Thursday	AM	0.44
Thursday	PM	0.56
Friday	AM	0.44
Friday	PM	0.56
Saturday	AM	0.37
Saturday	PM	0.63
Sunday	AM	0.37
Sunday	PM	0.63
Monday	AM	0.44
Monday	PM	0.56

Creel interview process

WDFW recreational crab creel sampling protocol for 2024 is briefly described below and in full detail in appendix A.

Upon arrival at a site, a sampling station was set up to easily view and intercept vessels as they pulled up to the ramp or other access points, while being positioned to avoid causing congestion due to sampling of fishers. During each contact, samplers identified themselves as WDFW staff conducting interviews to collect data on their fishing trip. At least one harvester from each vessel was asked by WDFW creel staff whether they had been crabbing (a visual assessment of any crab gear was used to confirm their response). The interview would proceed once it was confirmed that the intercepted vessel had been crabbing regardless of whether their crabbing trip was successful or not. Interviews were only conducted on completed trips for motorized and non-motorized (e.g., kayak, canoe, inflatable, etc.) crabbing vessels.

WDFW creel staff used an electronic survey form (*iForms*) to record data from harvesters during interviews. The survey form was designed to capture trip information from each vessel and compliance associated with each harvester's Catch Record Card and all crab in possession onboard the vessel. After arriving at a site, a sampler immediately recorded the date, time, and location. Once an interview had been initiated, the vessels trip information was recorded, which included the time the interview was conducted, the number of individuals that engaged in crabbing on the vessel, the number of pots and other types of gear (ring nets, star traps, snares) fished, the marine area and creel area (e.g., sub-area) their gear was set in, the number of lost pots or other gear, the total number of Dungeness crab retained on the vessel, and the total number of red rock crab retained on the vessel. Creel staff provided a map to the interviewee when asked what creel area their gear was set in. If Dungeness crab were possessed on the boat, the interview would proceed to ask all individuals that possessed crab and were responsible for recording their catch to show their CRC to WDFW creel staff to confirm recording compliance of their current crabbing trip. Samplers then recorded the number of CRCs that were checked, and the total number of Dungeness crab recorded across all CRCs. If no Dungeness crab were possessed, CRCs were not checked. After checking each harvester's CRC and documenting the number of crabs in possession, detailed biological and compliance information was recorded from each crab on the boat. Each sampler was equipped with a standard crab gauge that could accurately measure both 6 ¼ inches and 5 inches, and a bucket and scale that was calibrated prior to use in the field to weigh samples to the nearest 0.01 pounds. Due to time constraints, weight measurements and catch compliance of Dungeness and red rock crab were taken on the catch of every 3rd vessel encountered at a minimum. Harvesters were provided buckets by WDFW creel staff and were asked to put their catch into the buckets to be sampled. Creel staff recorded the total number of Dungeness or red rock crabs being checked on the vessel, the total number of female Dungeness crabs

possessed, the total number of sub-legal male Dungeness crabs (below 6.25 inches across the carapace) possessed, the total number of soft-shell male Dungeness crabs possessed, the number of female red rock crabs possessed, and the number of sub-legal red rock crabs (below 5 inches across the carapace) possessed. The method used to determine if a Dungeness crab was soft-shell is described in Appendix B and follows the same legal definition as used in all Puget Sound Dungeness crab fisheries. Weight measurements of Dungeness and red rock crab were taken in batches, recording the total number of crabs weighed and the total weight measured across all crabs on the boat. At the end of the sampling day, the end time of the creel was recorded, and the survey form was uploaded to the *iForms* database.

Limited sampling was conducted at a few select public fishing piers in close proximity to boat ramps when possible. It followed the same general sampling protocol as described above for boat ramps. In contrast to sampling protocols at boat ramps, creel interviews at piers included both completed and uncompleted trips. The pier survey was designed to be a quick ‘sweep’ of the pier to check catch compliance, and the survey would finish once every crabber on the pier was interviewed.

Where possible, any group or person that was participating in the Puget Sound recreational crab fishery out of accordance with the rules and regulations was informed of their non-compliance and educated. If a harvester was new to the fishery or was not familiar with the rules and regulations, they were given a variety of educational materials (See Appendix A for list of educational materials passed out).

Mean Catch per Harvester Trip

For the purpose of this report, we examine the average Dungeness crab caught and retained per individual harvester’s trip to better understand recreational success. For each boat contacted, the average catch per harvester was calculated by dividing the total number of Dungeness crab possessed by the total number of individuals engaged in crabbing for each vessel contacted. The mean Dungeness crabs caught and retained per individual harvester’s trip by MA is calculated using the following equation:

$$\text{Mean Dungeness per harvester trip by Marine Area} = \frac{\sum_{i=1}^n \left(\frac{a}{b} \right)_{i,d}}{n_{i,d}}$$

a = Number of Dungeness crabs retained by boat i in Marine Area d

b = Number of crab harvesters on boat i in Marine Area d

n = Number of recreational crab boats contacted in Marine Area d

Descriptive statistics and analysis

The results below describe patterns and observations from the summer 2024 Puget Sound recreational crab fishery. We first present and analyze data from the stratified random survey in MAs 8-1 and 8-2, followed by a brief summary of data collected in all other MAs through opportunistic sampling, and a brief summary of data collected at piers. All non-vessel contacts ($n = 2$) are excluded from summaries and analyses due to the small sample size and underrepresentation of this group. Furthermore, any analysis focused on assessments of catch compliance or weight measurements excludes catch where only a portion of the catch was sampled due to the inability to randomly sample only a portion of a vessels catch. Lastly, vessels contacted that were in possession of Dungeness crab, but CRCs were not checked for reasons other than non-compliance were removed from calculations of percent unrecorded catch. All analyses were performed in R (version 4.0.3; R Core Team 2020) and figures using the R package 'ggplot2'.

Results

MAs 8-1 and 8-2 Focus Area Survey

Sampling frequency

During the summer 2024 crab season, a total of 215 individual creels were conducted across 17 sites in MAs 8-1 and 8-2 (Table 4; Figure 3). Of these creels, 81 took place during the morning shift (AM) and 134 during the afternoon shift (PM), randomly selected based on probability weights. Across all 17 sites, we recorded 3,400 vessel contacts (excluding 2 non-vessel contacts) and checked 6,230 CRCs. Across these 3,400 vessel contacts, a total of 26,724 Dungeness crab were observed in possession. Of these, 51% (13,558 crabs) were assessed for compliance data, and 39% (10,481 crabs) were weighed, excluding any sub-samples taken. Additionally, we observed 1,422 red rock crabs in possession; 43% (610 crabs) of which were checked for compliance data, and 31% (440 crabs) were weighed.

The five most frequently sampled sites included: Everett 10th St. boat ramp (n = 61 creels), Cornet Bay boat ramp (n = 32 creels), Camano Island State Park boat ramp (n = 23 creels), Kayak Point boat ramp (n = 17 creels), and Oak Harbor boat ramp (n = 15 creels). Collectively, these five sampling locations accounted for 83% of the total number of vessels contacted (Figure 3), 81% of the CRCs checked, and 79% of the Dungeness and 85% of the red rock crab sampled for compliance and biological data during the MA 8-1 and 8-2 stratified survey.

Contact types

Of the 3,402 total contacts in MAs 8-1 and 8-2, 3,353 were motorized vessels, 47 were non-motorized vessels (e.g., kayak, canoe, paddleboard, etc.), and 2 were non-vessel contacts (1 group of scuba divers and 1 intertidal wader). Not all vessels contacted across the 17 sites set crab gear in MAs 8-1 or 8-2 (Table 5). Of the 3,400 vessel contacts, 166 did not set crab gear in MAs 8-1 or 8-2 during their trip. Cornet Bay ramp had the greatest variability in areas fished, with 65% in 8-1, 15% in 8-2, 15% in 6, and 4% in 7S. Boats utilizing the other sites fished primarily in either 8-1 or 8-2.

Table 4. Summary of creel sampling frequency across the 17 sites surveyed in MAs 8-1 and 8-2 in 2024. Dungeness sampled and red rock crab sampled refers to the number of crabs sampled for compliance (i.e., checking for female, sub-legal, and soft-shell) and excludes catch from vessels where only a portion of the total catch was sampled. This table excludes data from two non-vessel contacts. AM shifts occurred between 08:00 and 14:00. PM shifts occurred between 14:00 and 20:00.

Creel location	AM Shifts	PM Shifts	Vessels contacted	CRCs Checked	Dungeness sampled: compliance	Red rock sampled: compliance
Everett 10 th St Ramp	26	35	1,692	2,874	5,708	231
Cornet Bay Ramp	13	19	492	922	1,745	138
Camano Island SP Ramp	8	15	298	609	1,474	122
Kayak Point Ramp	8	9	226	357	1,027	23
Maple Grove Ramp	4	9	168	355	978	7
Oak Harbor Ramp	5	10	117	283	786	2
Mukilteo Ramp	4	4	97	185	547	73
Everett Marina	4	8	80	125	368	5
Oak Harbor Marina	2	4	76	240	185	0
Utsalady Ramp	2	4	75	160	348	0
Cavalero County Park Ramp	0	3	25	32	83	1
Captain Coupes Park Ramp	0	3	19	35	135	0
La Conner Ramp	1	3	13	20	73	3
Ebey Waterfront Ramp	1	4	11	15	52	0
Langley Ramp	1	1	7	9	32	3
Langus Riverfront Park Ramp	2	1	2	6	6	2
Freeland County Park Ramp	0	2	2	3	11	0
Total	81	134	3,400	6,230	13,558	610

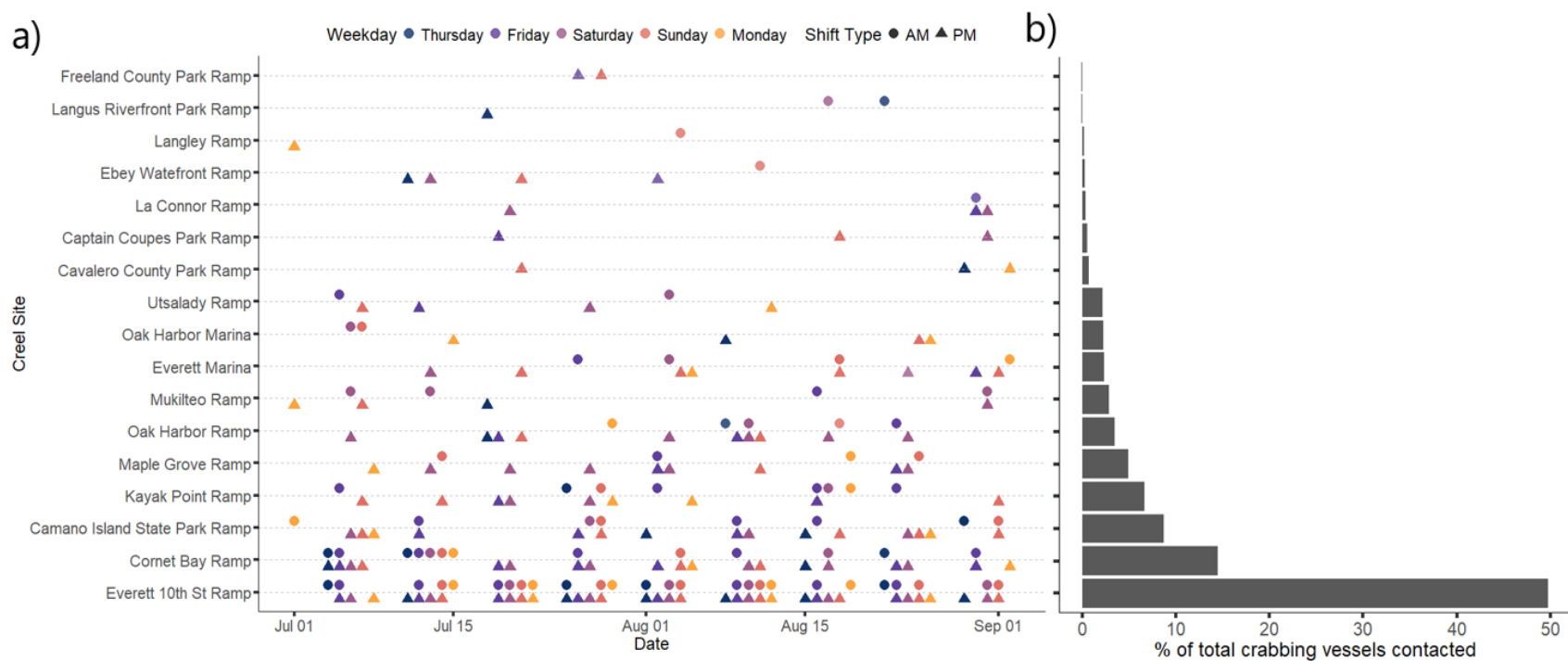


Figure 3. The number of 6-hour creels conducted throughout the 2024 summer crab season (date; x-axis figure a) and percent of the total crabbing vessels contacted (x-axis figure b) at each creel site (y-axis). In figure a, each point corresponds to an individual 6-hour creel conducted by WDFW sampling staff. Color represents the day of the week the creel was conducted, and the shape of each point corresponds to the shift type selected. In figure b, each bar represents the percent of the total number of vessels contacted at each creel location.

Table 5. Total number of vessels contacted at each site by Marine Area of harvest. For a specific creel location, dark color shade indicates a larger proportion of vessels were observed fishing in that MA.

Creel Site	MA 6	MA 7S	MA 7N	MA 8-1	MA 8-2	MA 9	MA 10	MA 11	MA 12
Everett 10th St Ramp	1	0	0	57	1,578	49	6	0	1
Cornet Bay Ramp	76	19	0	321	76	0	0	0	0
Camano Island SP Ramp	0	0	0	148	149	1	0	0	0
Kayak Point Ramp	0	0	0	14	210	1	1	0	0
Maple Grove Ramp	0	0	0	103	65	0	0	0	0
Oak Harbor Ramp	0	0	0	95	22	0	0	0	0
Mukilteo Ramp	0	0	0	1	90	6	0	0	0
Everett Marina	0	1	0	0	78	1	0	0	0
Oak Harbor Marina	0	0	0	72	4	0	0	0	0
Utsalady Ramp	0	0	0	63	12	0	0	0	0
Cavalero County Park Ramp	0	0	0	0	25	0	0	0	0
Captain Coupes Park Ramp	0	0	0	18	1	0	0	0	0
La Connor Ramp	1	2	0	10	0	0	0	0	0
Ebey Waterfront Ramp	0	0	0	0	11	0	0	0	0
Langley Ramp	0	0	0	0	7	0	0	0	0
Freeland County Park Ramp	0	0	0	2	0	0	0	0	0
Langus Riverfront Park Ramp	0	0	0	0	2	0	0	0	0

Temporal variability in vessels contacted

The mean number of vessels returning to the dock and finished crabbing for the day varied greatly between sites (Figure 3). The Everett 10th St ramp recorded the highest average number of vessels contacted per creel hour, peaking between 12:00 and 16:00 with an average of approximately 6 vessels per creel hour. Some sites, such as the Cornet Bay ramp, exhibited more than one peak in the mean number of vessels returning to the dock throughout the day. The mean number of vessels contacted also varied based on the day of the week and the sampling site (Figure 4). Generally, afternoon shifts (PM shifts) had a higher average number of vessels contacted each weekday compared to morning shifts (AM shifts). While the average number of vessels contacted varied widely by site, it was typically highest on Saturdays and lowest on Thursdays. Notably, the Cornet Bay ramp maintained a relatively consistent average throughout the week but experienced a peak on Thursdays during the afternoon shift. Furthermore, morning and afternoon shifts at the Camano Island State Park ramp and afternoon shifts at the Kayak Point ramp on Mondays saw a higher-than-expected mean number of vessels contacted (Figure 4).

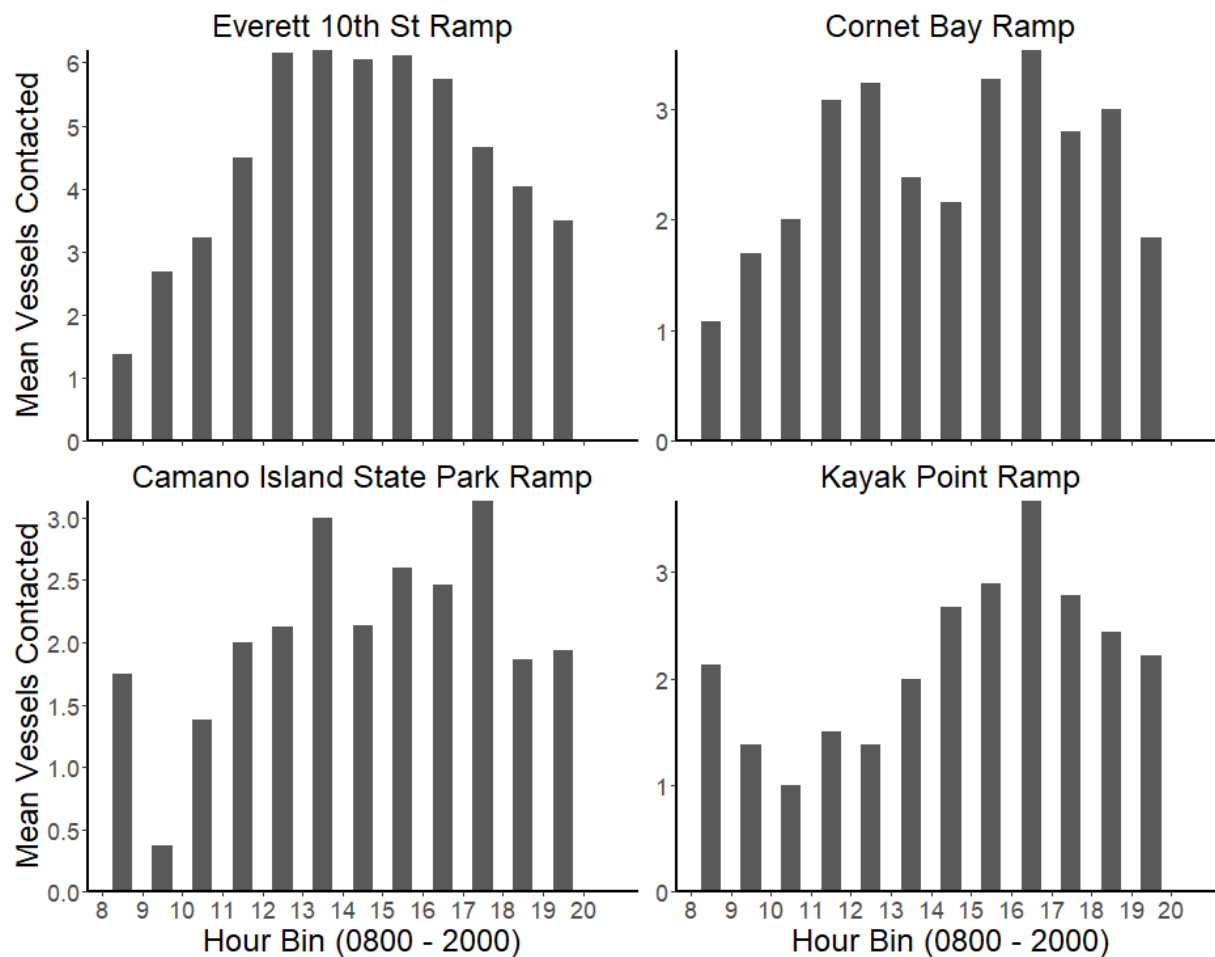


Figure 4. Mean number of vessels contacted at the four most frequently sampled sites by hour bin (2400-hour clock) for the summer 2024 recreational crabbing season.

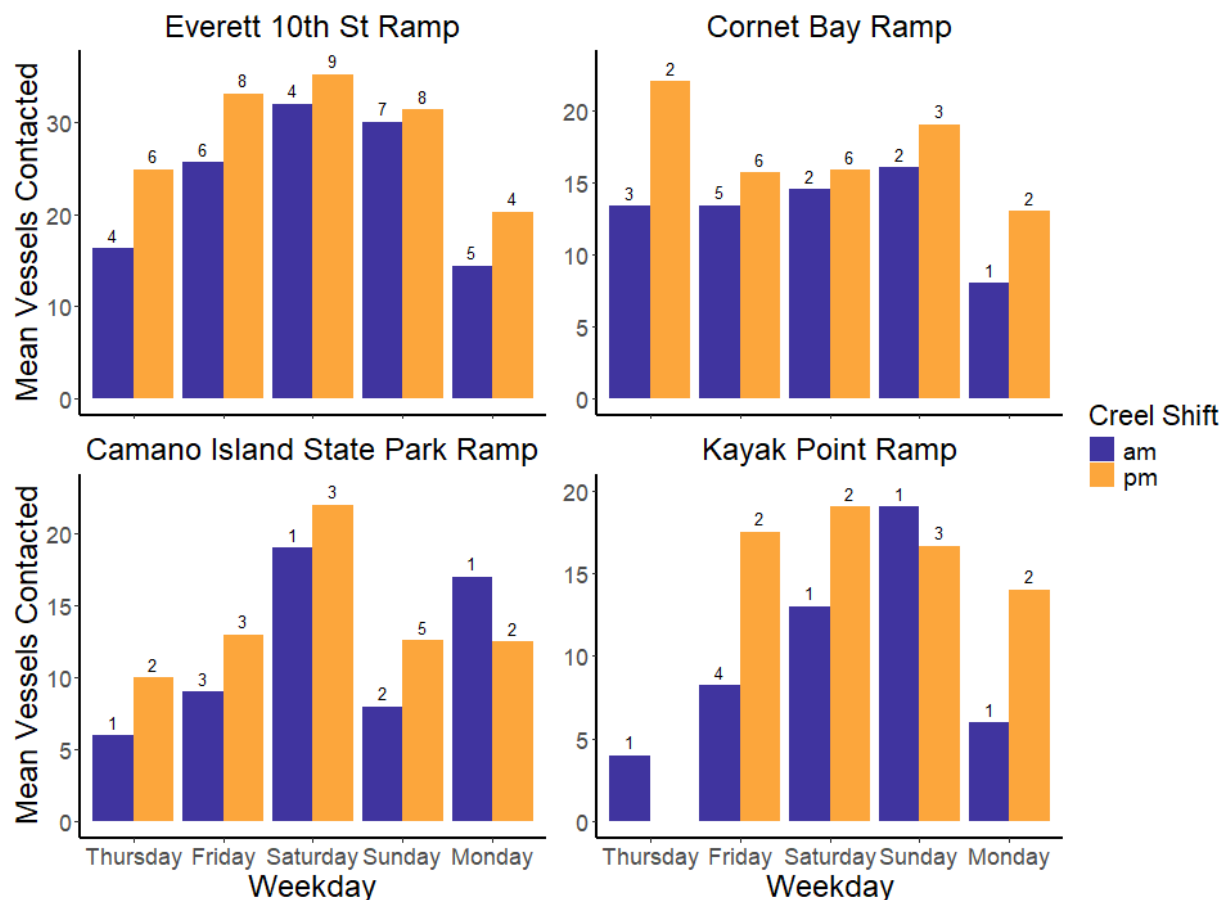


Figure 5. Mean vessels contacted at the four most frequently sampled sites by day of the week and shift strata. Numbers on top of each bar represent the number of creels conducted for a unique weekday and shift strata.

Compliance – Catch Record Cards (CRC) and unrecorded crab

A total of 3,076 vessels with Dungeness crab in possession were asked to show their summer Dungeness crab CRCs from all individuals that retained Dungeness crab and were responsible for recording their catch to WDFW creel staff. Of those vessels, 2% (n = 58) did not have any CRCs in possession and 0.5% (n = 15) did not comply with the request from WDFW creel staff to view CRCs for recording compliance. Overall, the proportion of vessels with no CRCs in possession and vessels that did not comply with the request from WDFW creel staff to view their CRC was low and varied little between MAs 8-1 and 8-2 (Table 6).

Table 6. Total number of vessels contacted, vessels with Dungeness crab in possession, and CRCs checked by creel staff, percent of vessels contacted with no CRCs in possession, and percent of vessels that did not comply with the request to view CRCs.

Marine Area of harvest	Vessels contacted	Vessels w/ Dungeness retained	CRCs checked	% vessels with no CRCs in possession	% vessels non-compliant with CRC request
8-1	904	873	1,905	1%	0.5%
8-2	2,330	2,203	3,994	2%	0.5%

Recreational harvesters of Dungeness crab in Puget Sound are required to immediately record each Dungeness crab kept onto their CRC when landed. The number of unrecorded Dungeness crab was determined by comparing the total number of Dungeness crab retained on a vessel to what was recorded across all CRCs on the vessel at the time the vessel was contacted, and the interview began. Of the 8,519 Dungeness crab caught and retained in MA 8-1, 13.48% (1,148 crabs) were not recorded on CRCs at the time of the interview (Table 7). In MA 8-2, 16.00% (2,650 crabs) of the 16,558 Dungeness caught and retained by recreational harvesters were not recorded on CRCs at the time of the interview.

The creel efforts described here do not capture instances where recreational harvesters wait to record their catch until off the water at their residence or another location. During recreational crab creels in 2023, we calculated the proportion of the successful recreational harvester trips that were not recorded on a CRC at the time of the creel interview but were later reported to WDFW. This was calculated by searching the CRC database using a harvester's document identification number to find if a harvester's reported trip matched the date and MA of harvest indicated during the creel interview. This method found that, despite many unrecorded crabbing trips observed at the time of the creel interview, 56% of those total unrecorded crabbing trips went on to be reported to the department. These results may indicate that many recreational harvesters wait to record their catch until a later time. Alternatively, it may also indicate the presence of our creel staff had a positive influence on whether harvesters record and report their catch at a later time.

Table 7. The number and percent of Dungeness crabs retained by recreational harvesters that were not recorded on Catch Record Cards (CRCs) prior to the time of the interview conducted by WDFW creel staff. The percent of Dungeness crab recorded on summer CRCs is calculated as the total number of unrecorded Dungeness crab divided by the total number of Dungeness crab retained. Vessels contacted that were in possession of Dungeness crab, but CRCs were not checked for reasons other than non-compliance were removed from calculations of percent unrecorded catch.

Marine Area of harvest	Total Dungeness retained	Dungeness recorded on CRCs	Dungeness unrecorded on CRCs	% Dungeness unrecorded on CRCs
8-1	8,519	7,371	1,148	13.48%
8-2	16,558	13,908	2,650	16.00%

* *An unrecorded Dungeness crab is defined as a Dungeness crab in possession by a recreational harvester that is not recorded on their Catch Record Card at the time the creel interview was conducted.*

Compliance – Dungeness crab catch

Female Dungeness crab. Retention of female Dungeness crab was rare, representing only 0.5% (17 females) of the total number of Dungeness crab sampled for compliance and biological data in MA 8-1 and only 0.3% (30 females) in MA 8-2 (Table 8).

Sub-legal male Dungeness crab. Of the 3,767 Dungeness crab sampled for compliance and biological data in MA 8-1, 3.1% (116 sub-legal) were observed to be below the minimum size limit of 6.25 inches (Table 8). Retention of sub-legal Dungeness crab had a higher frequency in MA 8-2, of which 6.3% (581 sub-legal) of the Dungeness crab sampled were below the legal-size limit (Table 8).

Soft-shell male Dungeness crab. Retention of soft-shell Dungeness crab appeared to be the most frequent infraction. Soft-shell Dungeness crab in MA 8-2 represented 10.5% (963 soft-shell) of the total number of Dungeness crab sampled (Table 8). Of the 3,767 Dungeness crab sampled for compliance and biological data in MA 8-1, 4.3% (162 soft-shell) were observed to be soft-shell.

Compliance – red rock crab catch

Of the 584 red rock crab sampled for compliance and biological data across MAs 8-1 and 8-2, 9.2% (n = 54) were observed to be below the minimum size limit of 5 inches. Possession of sub-legal red rock crab was most frequently observed in MA 8-2, where 10.5% of the sampled red rock crab were below the minimum size limit of 5 inches (Table 8). Female red rock crab, although legal to retain in the recreational fishery, represented 6.3% (n = 37) of the total number of red rock crab sampled for biological data, indicating that male crabs represent the majority of retained red rock crab in the recreational fishery.

Table 8. Sample size and percent occurrence of retained female, sub-legal, and soft-shell Dungeness and red rock crab by recreational harvesters in MAs 8-1 and 8-2. The percent of female, sub-legal, and soft-shell crab, indicated in '()', is expressed as the total number of females, sub-legal, or soft-shell crab sampled from recreational harvesters within a MA divided by the total number of Dungeness or red rock crab sampled within a MA. Red rock crabs were not assessed for soft-shell. Catch from vessels where only a portion of the total catch was sampled is excluded.

Species	Marine Area of harvest	Total Crab sampled	Female crab: <i>n</i> observed and (% of total)	Sub-legal male crab: <i>n</i> observed and (% of total)	Soft-shell male crab: <i>n</i> observed and (% of total)
Dungeness Crab	8-1	3,767	17 (0.5%)	116 (3.1%)	162 (4.3%)
Dungeness Crab	8-2	9,190	30 (0.3%)	581 (6.3%)	963 (10.5%)
Red Rock Crab	8-1	185	3 (1.6%)	12 (6.5%)	-
Red Rock Crab	8-2	399	34 (8.5%)	42 (10.5%)	-

Catch and effort patterns

The mean number of harvesters per vessel and the mean number of units of gear fished per vessel were calculated for each MA (Figure 5; Table 9). The mean number of harvesters and units of gear fished varied little among MAs 8-1 and 8-2. Despite little variation between the MAs, the highest estimated mean number of harvesters and units of gear fished per vessel was observed in MA 8-1 (2.88 harvesters; 4.10 units of gear fished).

Of the 3,234 vessels contacted across both MAs 8-1 and 8-2, 4.9% (158 vessels) were unsuccessful in catching and retaining any Dungeness crab during their trip. By MA, 3.4% of vessels (31 vessels) that fished in MA 8-1 were unsuccessful in catching and retaining any Dungeness crab, whereas 5.5% (127 vessels) were unsuccessful in MA 8-2. Catch rates tended to be the highest in MA 8-1 compared to MA 8-2 (Figure 5; Table 9). The mean Dungeness crabs retained per vessel ranged from 7.15 in MA 8-2 to 9.49 in MA 8-1, whereas the mean Dungeness crabs retained per individual harvester trip ranged from 2.70 Dungeness crabs in MA 8-2 to 3.32 Dungeness crabs in MA 8-1.

Red rock crab represented 4.9% (*n* = 1,294 crabs) of the total recreational crab retained throughout the creel survey in MAs 8-1 and 8-2. By MA, 4.9% of the total retained crab in MA 8-1 was comprised of red rock crab and 4.8% in MA 8-2 (Figure 6). The ratio of red rock crab retained to the number of Dungeness crab retained can be calculated by dividing the total number of red rock crab retained across all boats by the total number of Dungeness crab retained. In both MAs 8-1 and 8-2 the ratio of red rock crab retained to Dungeness crab

retained ranges from 0.051 (MA 8-2) to 0.052 (MA 8-1), indicating that 1 red rock crab is harvested for approximately every 19 Dungeness crab (Table 9).

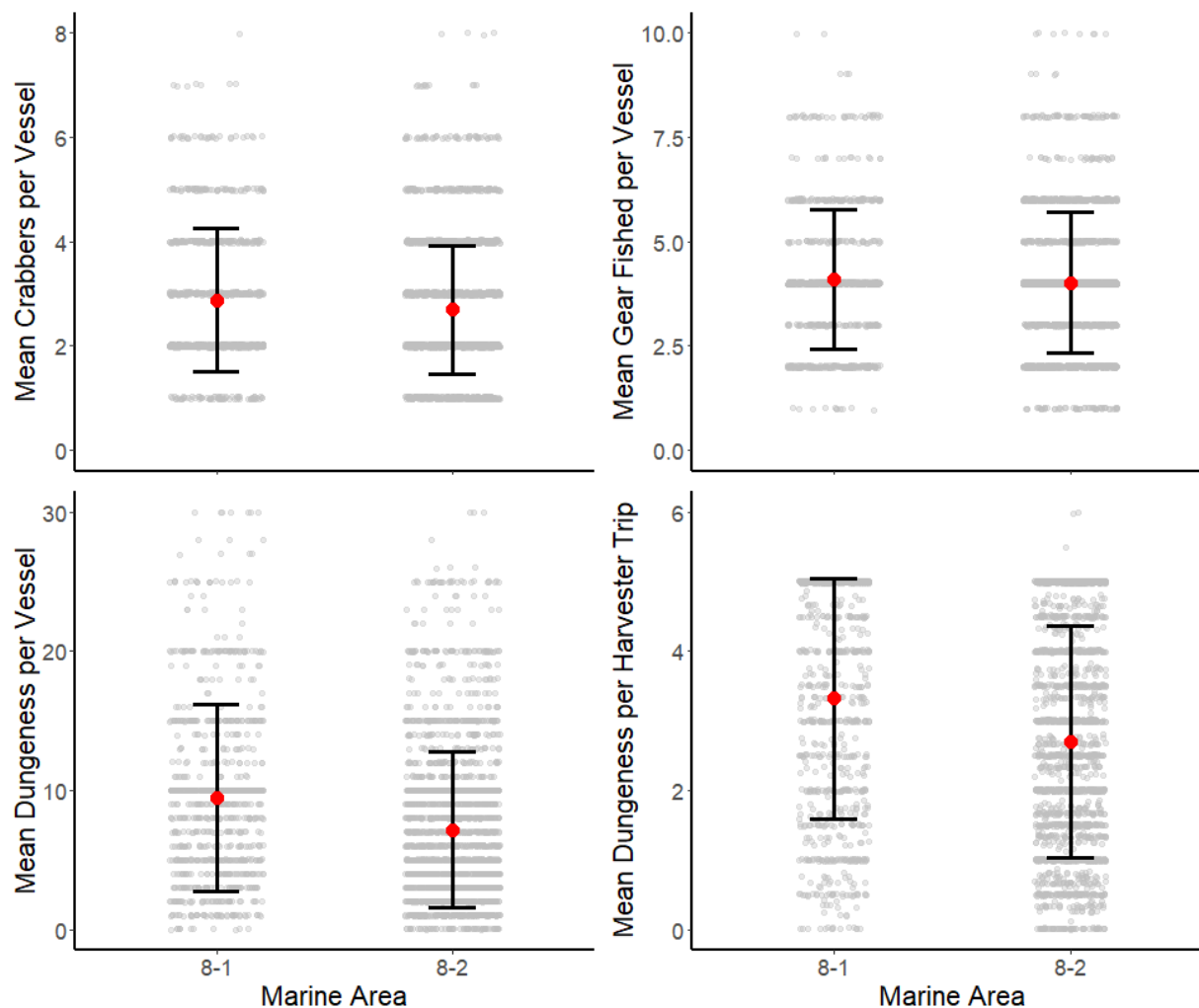


Figure 6. Interval plots showing mean crabbers per vessel, mean units of gear fished per vessel, mean number of Dungeness caught and retained per vessel, and mean number of Dungeness caught and retained per individual harvester trip by MA (x-axis). A harvester is defined as a crabber on board the interviewed vessel that was engaged in crabbing. A unit of gear includes pots, ring nets, and star traps. Red points indicate the mean, gray points indicate observed values, and vertical black lines above and below the mean represent $\pm$ standard deviation.

Table 9. Summary of the mean number of harvesters per vessel, gear fished per vessel, Dungeness caught and retained per vessel, Dungeness caught and retained per individual harvester trip, and the proportion and ratio of red rock crab retained to Dungeness crab by MAs 8-1 and 8-2. The standard deviation associated with each mean value is indicated in '()'. A harvester is defined as a crabber on board the contacted vessel that was engaged in crabbing. A unit of gear includes pots, ring nets, and star traps. Red rock crab expressed as the percent of total crab retained is calculated as the total number of red rock crab retained by recreational harvesters divided by the sum of the total number of red rock crab and Dungeness crab retained. The catch ratio is calculated as the total number of red rock crab observed divided by the total number of Dungeness crab observed.

Marine Area of harvest	Mean harvesters per vessel	Mean gear fished per vessel	Mean Dungeness per vessel	Mean Dungeness per harvester trip	Red rock crab as % of total crab retained	Catch ratio: Red Rock crab to Dungeness crab
8-1	2.88 (± 1.38)	4.10 (± 1.68)	9.49 (± 6.72)	3.32 (± 1.73)	4.9%	0.052
8-2	2.69 (± 1.24)	4.01 (± 1.69)	7.15 (± 5.60)	2.70 (± 1.67)	4.8%	0.051

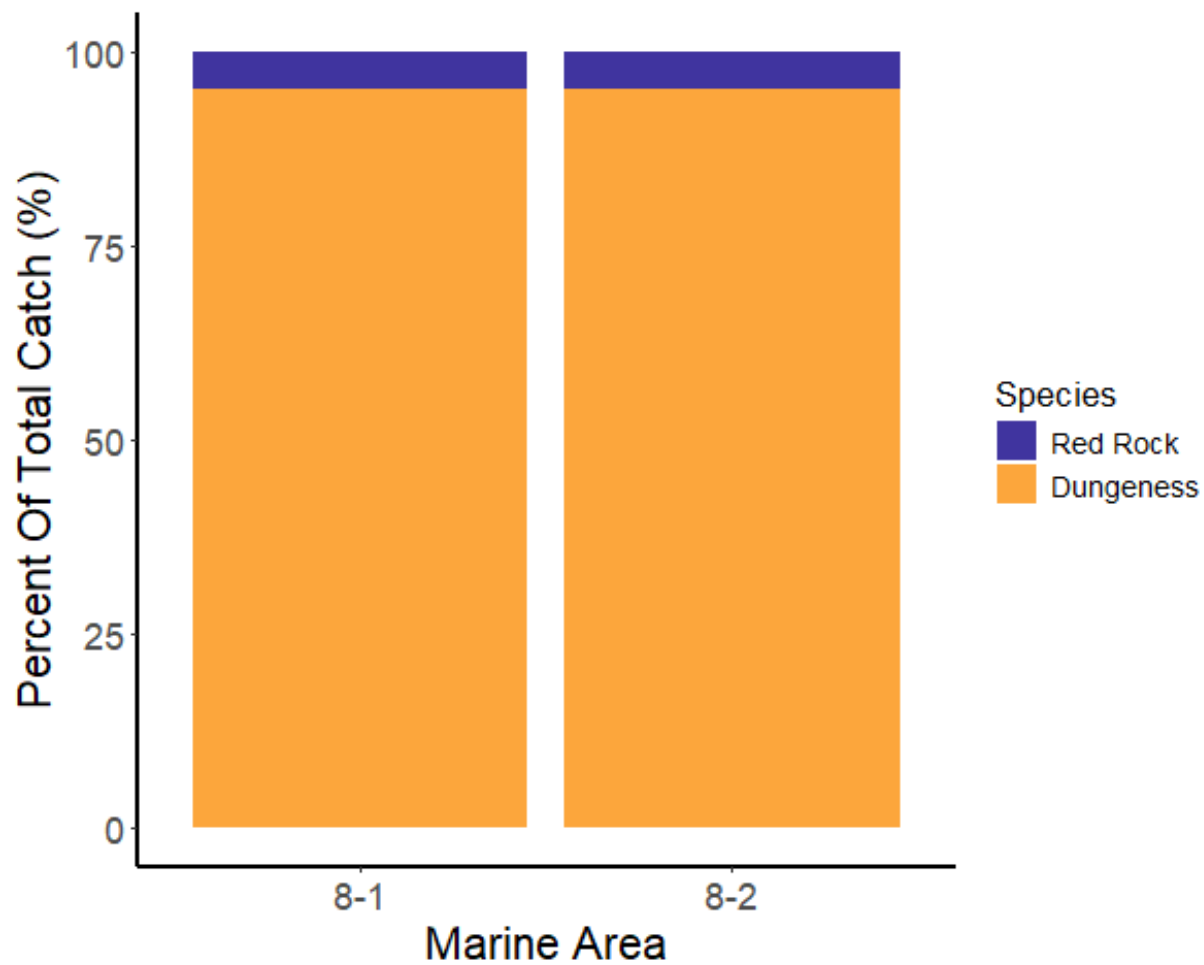


Figure 7. Dungeness and red rock crab expressed as percent of total crab retained by recreational harvesters by MA.

Weight measurements by marine area

Dungeness crab. Weight measurement data (lbs.) were taken from 401 vessels in MA 8-1 and 1,142 vessels in MA 8-2, excluding those where only a portion of the catch was weighed. A total of 2,952 Dungeness crabs in MA 8-1 and 7,041 in MA 8-2 were sampled by collecting a batch weight by weighing all crabs on the vessel together and noting the total number of crabs weighed. A mean weight per individual Dungeness crab was calculated by dividing the total pounds of Dungeness crab by the total number of individual Dungeness crabs weighed across all vessels. The mean pounds per Dungeness crab varied among the MAs (Table 10). The lightest crabs on average were found in MA 8-2 (1.75 mean lbs. per crab), whereas the heaviest crabs on average were found in MA 8-1 (1.88 mean lbs. per crab).

Red rock crab. Weight measurement data (lbs.) were taken on red rock crab from 46 vessels in MA 8-1 and 149 vessels in MA 8-2, excluding those where only a portion of the catch was weighed. A total of 133 red rock crabs in MA 8-1 and 283 in MA 8-2 were sampled by collecting a batch weight by weighing all crabs on the vessel together and noting the total number of crabs weighed. A mean weight per individual red rock crab was calculated by dividing the total pounds of red rock crab weighed by the total number of individual red rock crab weighed across all vessels. Red rock crab taken in the recreational fishery were much lighter compared to Dungeness crab, with the mean pounds per individual red rock crab ranging from 0.96 lbs. in MA 8-1 to 1.04 lbs. in MA 8-2 (Table 10).

Table 10. Sample size of the total number of crabs weighed, total pounds observed, and mean pounds per individual Dungeness and red rock crab by MAs 8-1 and 8-2. The mean weight per crab for each MA was calculated by dividing the total pounds by the total number of individual crabs weighed. The standard error around the mean weight per crab was estimated using a bootstrap procedure ($N = 2,000$). Catch from vessels where only a portion of the total catch was weighed is excluded.

Species	Marine Area of harvest	Total # of crab weighed	Total Pounds weighed (lbs.)	Mean weight per crab (lbs.)	Bootstrap std. error
Dungeness crab	8-1	2,952	5,551.76	1.88	0.02
Dungeness crab	8-2	7,041	12,325.19	1.75	0.01
Red rock crab	8-1	133	128.19	0.96	0.05
Red rock crab	8-2	283	293.16	1.04	0.03

Lost recreational crab gear

Each vessel contacted is asked how many units of gear (pots, ring nets, or star traps) were lost during their crabbing trip. A total of 164 units of gear were reported lost to WDFW crab creel staff across the 3,234 vessels that set crab gear in MAs 8-1 and 8-2 (Table 11). By MA, 2.8% (25 vessels) of the vessels contacted in MA 8-1 reported lost gear during their crabbing trip, whereas 5% (115 vessels) of the contacted vessels in MA 8-2 lost gear. In MA 8-1, 34 units of gear were reported lost to WDFW creel staff, which comprised 0.9% of the total gear fished across all contacted vessels. In MA 8-2, the rate of gear loss was higher, with 115 units of crab gear reported lost, comprising 1.4% of the total gear fished across all contacted vessels.

Table 11. Summary of recreational crab gear loss, including the total number and percent of vessels contacted that reported at least one unit of gear lost during their trip, total units of gear reported lost, total units of gear that had been fished across all contacted vessels in each MA, and the percent of gear fished that was reported lost. A unit of crab gear is defined as either a crab pot, ring net, or star trap.

Marine Area of harvest	Total contacted vessels reporting lost gear	% contacted vessels reporting lost gear	Total units of gear reported lost	Total units of gear fished	% of gear fished reported lost
8-1	25	2.8%	34	3,702	0.9%
8-2	115	5.0%	130	9,347	1.4%

General Compliance Survey in all other MAs

Sampling conducted opportunistically, outside of the MA 8 stratified survey, consisted of 103 individual creels (n = 577 creel hours) across 27 sites (Table 12). Across all 27 sites, 1,310 vessel contacts were made, and 2,282 CRCs were checked. In terms of Dungeness crab, a total of 9,268 were observed in possession. Out of these, 56% (5,145 crabs) were assessed for compliance data, and 33% (3,053 crabs) were weighed, excluding any sub-samples taken. Additionally, 2,940 red rock crabs were observed in possession; 38% (1,109 crabs) of these were checked for compliance data, and 21% (616 crabs) were weighed.

The five most frequently sampled sites included: John Wayne Marina boat ramp (n = 106 creel hours), Point Defiance boat ramp (n = 70 creel hours), Salsbury Point County Park boat ramp (n = 46 creel hours), Shilshole boat ramp (n = 43 creel hours), and the Port Townsend Boat Haven ramp (n = 42 creel hours). Collectively, these five sites accounted for 70% of the total number of vessels contacted, 71% of the CRCs checked, and 71% of the Dungeness and 92% of the red rock crab sampled for compliance and biological data during the general compliance surveys. The Point Defiance boat ramp alone accounted for 65% of the total number of red rock crab checked for compliance and biological data.

Generally, each creel site was associated with a single MA of harvest (Table 13). Only creel sites with close proximity to the boundary of multiple MAs were associated with more than one MA of harvest. Outside the Cornet Bay boat ramp, the Salsbury Point County Park ramp had the greatest variability in areas fished, with 58% of vessels contacted setting crab gear in MA 9 and 42% in MA 12.

Table 12. Summary of creel effort (creel hours), number of vessels contacted, number of CRCs checked, and number of Dungeness and red rock crab sampled for compliance and biological data by creel locations sampled opportunistically during general compliance creels.

Creel Location	Creel Hours	Vessels Contacted	CRCs Checked	Dungeness Crab Checked	Red Rock Crab Checked
John Wayne Marina Ramp	106	350	768	1,814	112
Point Defiance Ramp	70.3	328	467	884	716
Salsbury Point County Park Ramp	45.9	67	88	207	151
Shilshole Ramp	42.9	106	171	470	38
Port Townsend Haven Ramp	42.1	63	136	263	4
Port Angeles West Ramp	26.6	24	44	25	0
Everett 10th St Ramp	26	86	99	247	7
Cline Spite County Park Ramp	24.8	36	55	125	3
Cap Sante Marina	23	51	121	207	9
Seabeck Ramp	19.2	15	15	61	19
Blaine Harbor Ramp	17.7	45	83	197	1
Squalicum Harbor Ramp	16.4	26	48	51	0
Washington Park Ramp	15.4	11	22	60	1
Kingston Marina Ramp	13.8	21	31	116	7
Cornet Bay Ramp	13.1	12	21	65	3
Ebey Waterfront Ramp	12.7	2	5	23	0
Birch Bay Ramp	8.9	15	27	92	0
Port Townsend Haven Marina	8.8	1	4	20	0
Kayak Point Ramp	8	16	19	68	0
Don Armeni Ramp	8	20	29	74	25
La Connor Ramp	6	7	9	25	13
Quilcene Ramp	5.5	0	0	0	0
Pleasant Harbor Marina	4.8	0	0	0	0
Birch Bay Beach	4.1	3	8	33	0
Dungeness Landing Park Ramp	3.5	4	10	18	0
Edmonds Boat Sling	2	1	2	0	0
Langus Riverfront Park Ramp	2	0	0	0	0
Total	577.4	1,310	2,282	5,145	1,109

Table 13. Total number of vessels contacted at each site by Marine Area of harvest. For a specific creel location, dark color shade indicates a larger proportion of vessels fished in that MA. Only creel locations where at least 10 vessels were contacted are included.

Creel Location	MA 6	MA 7S	MA 7N	MA 8-1	MA 8-2	MA 9	MA 10	MA 11	MA 12
John Wayne Marina Ramp	350	0	0	0	0	0	0	0	0
Point Defiance Ramp	0	0	0	0	0	1	2	325	0
Shilshole Ramp	0	0	0	0	1	1	104	0	0
Everett 10th St Ramp	0	0	0	0	83	3	0	0	0
Salsbury Point Park Ramp	0	0	0	0	0	39	0	0	28
Port Townsend Haven Ramp	2	0	0	0	0	61	0	0	0
Cap Sante Marina	0	51	0	0	0	0	0	0	0
Blaine Harbor Ramp	0	0	45	0	0	0	0	0	0
Cline Spite County Park Ramp	36	0	0	0	0	0	0	0	0
Squalicum Harbor Ramp	0	23	3	0	0	0	0	0	0
Port Angeles West Ramp	24	0	0	0	0	0	0	0	0
Kingston Ramp	0	0	0	0	0	4	17	0	0
Don Armeni Ramp	0	0	0	0	0	0	19	1	0
Kayak Point Ramp	0	0	0	0	16	0	0	0	0
Birch Bay Ramp	0	0	15	0	0	0	0	0	0
Seabeck Ramp	0	0	0	0	0	0	0	0	15
Cornet Bay Ramp	2	1	0	9	0	0	0	0	0
Washington Park Ramp	0	11	0	0	0	0	0	0	0

Pier sampling

Pier sampling was conducted on a limited basis. During the Puget Sound summer 2024 recreational crab season, a total of 41 individual creels were conducted across ten piers in six different MAs (Table 14). Across all ten piers sampled, 178 pier harvesters (or harvester groups) were contacted, and 62 CRCs were checked for individuals that had Dungeness crab in their possession. Of the 110 Dungeness crabs observed in possession across all pier locations, 46% (51 crabs) were not recorded on a Puget Sound summer CRC (Table 15). Of the 94 total Dungeness crab sampled, 29 Dungeness crabs were under the minimum size limit of 6.25 inches, 35 Dungeness crabs were observed to be soft-shell, and three females were observed in possession (Table 16). Among the ten piers, 239 red rock crab were sampled of which 28 were female (legal to retain female red rock crab), 42 were below the minimum size limit of 5 inches, and 26 soft-shell red rock crabs were observed in possession.

Table 14. Sampling frequency of recreational harvesters at pier locations, number of harvesters (or harvester groups) interviewed, and number of Catch Record Cards (CRCs) checked during the summer 2024 Puget Sound recreational crab season. Only harvesters with Dungeness crab in possession were asked to view their CRC.

Pier location	Marine Area	Creels conducted	Interviews conducted	CRCs checked
Cornet Bay Fishing Pier	8-1	11	47	15
Mukilteo Fishing Piers	8-2	9	55	28
Edmonds Fishing Pier	9	5	29	11
Everett Marine Park Pier	8-2	4	7	1
Shilshole Pier	10	4	9	0
Port Angeles Fishing Pier	6	2	6	5
Seacrest Pier	10	2	5	0
Les Davis Fishing Pier	11	2	12	1
Port Townsend Fishing Piers	9	1	1	1
Des Moines Fishing Pier	11	1	7	0
Total	-	41	178	62

Table 15. The number of Dungeness crabs retained, recorded, and unrecorded on Catch Record Cards (CRCs) prior to the time of the interview by recreational harvesters at piers.

Pier Location	Total Dungeness retained	Dungeness recorded on CRCs	Dungeness unrecorded on CRCs
Cornet Bay Fishing Pier	26	13	13
Mukilteo Fishing Piers	52	27	25
Edmonds Fishing Pier	25	14	11
Everett Marine Park Pier	2	2	0
Shilshole Pier	0	-	-
Port Angeles Fishing Pier	3	3	0
Seacrest Pier	0	-	-
Les Davis Fishing Pier	1	0	1
Port Townsend Fishing Piers	1	0	1
Des Moines Fishing Pier	0	-	-
Total	110	59	51

Table 16. Sample size and percent occurrence of retained female, sub-legal, and soft-shell Dungeness and red rock crab by recreational harvesters at each pier location. The percent of female, sub-legal, and soft-shell crab, indicated in '()', is expressed as the total number of females, sub-legal, or soft-shell crab sampled from recreational harvesters at a pier location divided by the total number of Dungeness or red rock crab sampled.

Species	Pier location	Total crab sampled	Female crab: <i>n</i> observed and (% of total)	Sub-legal crab: <i>n</i> observed and (% of total)	Soft-shell crab: <i>n</i> observed and (% of total)
Dungeness Crab	Cornet Bay Fishing Pier	25	2 (8%)	6 (24%)	1 (4%)
Dungeness Crab	Mukilteo Fishing Piers	49	1 (2%)	16 (33%)	25 (51%)
Dungeness Crab	Edmonds Fishing Pier	13	0 (0%)	6 (46%)	9 (69%)
Dungeness Crab	Everett Marine Park Pier	2	0 (0%)	1 (50%)	0 (0%)
Dungeness Crab	Shilshole Pier	0	-	-	-
Dungeness Crab	Port Angeles Fishing Pier	3	0 (0%)	0 (0%)	0 (0%)
Dungeness Crab	Seacrest Pier	0	-	-	-
Dungeness Crab	Les Davis Fishing Pier	1	0 (0%)	0 (0%)	0 (0%)
Dungeness Crab	Port Townsend Fishing Piers	1	0 (0%)	0 (0%)	0 (0%)
Dungeness Crab	Des Moines Fishing Pier	0	-	-	-
Red Rock Crab	Cornet Bay Fishing Pier	73	0 (0%)	12 (16%)	6 (8%)
Red Rock Crab	Mukilteo Fishing Piers	48	15 (31%)	12 (25%)	7 (15%)
Red Rock Crab	Edmonds Fishing Pier	42	8 (19%)	9 (21%)	4 (10%)
Red Rock Crab	Everett Marine Park Pier	0	-	-	-
Red Rock Crab	Shilshole Pier	21	3 (14%)	4 (19%)	1 (5%)
Red Rock Crab	Port Angeles Fishing Pier	6	0 (0%)	0 (0%)	0 (0%)
Red Rock Crab	Seacrest Pier	4	1 (25%)	0 (0%)	0 (0%)
Red Rock Crab	Les Davis Fishing Pier	9	1 (11%)	1 (11%)	2 (22%)
Red Rock Crab	Port Townsend Fishing Piers	1	0 (0%)	0 (0%)	0 (0%)
Red Rock Crab	Des Moines Fishing Pier	35	0 (0%)	4 (11%)	6 (17%)

Summary

The Puget Sound recreational crab creel survey conducted July 1, 2024, through September 2, 2024, focused on collecting catch and compliance data of recreational crab harvesters in MAs 8-1 and 8-2. The survey results demonstrate variability in harvester recording compliance, harvest rates, effort, and mean weight of Dungeness crab across the MAs and indicated where catch compliance might be lacking, and education and outreach should be focused. The survey also provides fisheries-related information on the recreational fishery of red rock crab, where data is limited across all areas in Puget Sound. Data collected from the summer 2024 season in MAs 8-1 and 8-2 helps lay the groundwork to continue the same survey design and data collection methods to be implemented in other areas in future years.

There were several positive results from the creel survey. Over 98% of vessels contacted had all Catch Record Cards in possession during creel interviews and over 99% of harvesters on vessels complied with the survey questions. Moreover, we gained valuable data about compliance challenges (e.g., unrecorded Dungeness crab, retention of sub-legal and soft-shell Dungeness crab) in MAs 8-1 and 8-2. We also note that compliance improved compared to data collected the prior summer season (2023).

Compliance with recording Dungeness crab onto a Catch Record Card immediately once retained continues to be a challenge in the recreational crab fishery in Puget Sound. This compliance challenge was evident in MAs 8-1 and 8-2, where 14.1% (8-1) and 16.6% (8-2) of all crab retained by recreational harvesters were not recorded on CRCs prior to the interview. Despite this, our creel efforts would not have captured instances where recreational harvesters wait to record their catch until off the water and at their residence and therefore provides an overestimate of unrecorded catch. Recording compliance should only improve with continued creel efforts, education, and reminders to record their catch immediately after retention of each Dungeness crab.

The Puget Sound recreational Dungeness crab fishery is one of the most popular sport fisheries in Washington state. On average, 219,032 Puget Sound crab endorsements are sold each year (2014 - 2023). Therefore, maintaining a presence by WDFW crab creel staff to monitor the fishery, collect catch and compliance data, and educate harvesters is extremely valuable. Further, due to the dynamic nature of the fishery, one season of creel data in MAs 8-1 and 8-2 is not sufficient to characterize the recreational Dungeness crab fishery in Puget Sound across all areas. Continuing to monitor the Puget Sound recreational Dungeness crab fishery each year, along with a focus on the remaining Marine Areas will enhance data quality and continue to improve management.

Appendix A

Appendix A. 2024 Puget Sound Recreational Crab Creel Sampling protocol.

1. **Before leaving the office:** Check that all your gear is organized and in possession, you have signed out an assigned/available vehicle, and your scales are calibrated using the calibration weights.
2. Be at your designated sampling site in a timely manner. If you are scheduled start your creel at a specific time, it is important you arrive at your site early to ensure you start your creel at your assigned time. Set up your station and have your gear ready at the best spot to easily view and intercept boats as they pull up to the ramp or other access point, but where it will not create congestion.
3. Only interview vessels and harvesters that **attempted to go crabbing AND are done crabbing for the day**. Do not interview boats and anglers that went out only to drop crab gear overnight.
 - You may encounter crabbers from kayak, paddleboard, scuba, or intertidal wading. Please treat these groups the same way you would treat a boat.
4. You can either use **paper forms** to collect data or the **iform app**.
5. Once you arrive at your creel location and your gear is ready, select the “**CRAB CREEL FORM V2**” (if using iForms) and complete the header fields:
 - a) **Assigned MA8 Survey?** Select **Yes** or **No**. Select ‘Yes’ if assigned to a location in MA8 with an AM or PM shift.
 - b) **Morning or Afternoon Shift?** Select **AM** or **PM**. Note: This question will not appear if you selected ‘No’ to question ‘5a’.
 - c) **Creel Date.** Select the date of your creel.
 - d) **Creel Start Time.** Enter the time your creel started.
 - e) **Creel Location.** Select your assigned creel location.
 - f) **Creeler Name.** Select your name.
 - g) **Weights in pounds or grams?** Select **Pounds, Grams, or No Scale**. Make sure to use the same scale units for each interview. Do not switch units throughout the day.
6. Intercept each boat at your location to request an interview after they have tied up and are proceeding to get their vehicle and trailer. Identify yourself as an employee of the state and WDFW and politely ask them if they have been crabbing today and to take a moment to answer some questions regarding their day crabbing. Be sure to explain what we are doing and why. Direct fishers to your sampling station or the best place to ask questions and record data. If there are multiple fishers on board, you can start the interview with one individual while the others are walking to get their vehicle and trailer.

7. **Boat/Trip information.** When you are ready to begin your first interview of the day, select the **COLLECT BOAT DATA** Subform field. The questions in this subform are focused on collecting trip and catch information from this boat, Catch Record Card recording compliance, catch compliance, and weight data.
- a) **Time of interview:** Select the time you started interviewing this boat.
 - b) **Is this a motorized vessel?** Select **Yes** or **No**. Select no if using a kayak, paddleboard, intertidal wading, scuba, etc. Note: If you select 'Other' a new field will appear where you can type in the method of harvest.
 - c) **Number of crabbers on this vessel?** Enter the number of crabbers in this party attempting to harvest crab. Note: This question still applies to non-motorized vessels or other methods of harvest.
 - d) **Number of crab pots fished?** Enter the number of crab **POTS** fished (including pots that were lost).
 - e) **Number of other units of crab gear fished?** Enter the number of other units of crab gear fished (including gear that was lost). This includes ring nets, star traps, and crab snares.
 - f) **Marine Area the crab gear was set in?** Select the Marine Area their crab gear was set in. Note: If setting gear in multiple Marine Areas, select the area the majority of their gear was set in.
 - g) **Catch Area the crab gear was set in?** Select the Catch Area their crab gear was set in. Note: If setting gear in multiple Creel Areas, select the area the majority of their gear was set in.
 - h) **Units of gear that were lost on this crabbing trip?** Enter the total units of gear that were lost from this trip.

Now, we need to record the number of Dungeness crab and red rock crab in possession on this boat. If it is difficult to view and count the crab or there are too many crabs to visually count where they are stored on the boat, please have the interviewees separate all catch into 5-gallon buckets. If any red rock crab is possessed, please separate red rock crab from Dungeness crab.
PLEASE DO NOT BOARD THE VESSEL.

After counting the number of Dungeness and red rock crab in possession, record the following:

- i) **Number of Dungeness crab in possession.** You must confirm this number by counting the total number of Dungeness crab in possession on this boat, do not take their word for it. Enter 0 if no Dungeness crab were kept.
- j) **Number of Red Rock crab in possession.** You must confirm this number by counting the total number of red rock crab in possession on this boat, do not take their word for it. Enter 0 if no red rock crab were kept.

Now you can check all Dungeness crab Catch Record Cards (CRC) and record the total number of Dungeness crab recorded. Fishers are required to have their CRC in possession if they harvest any Dungeness crab. Each Dungeness crab retained is required to be recorded immediately on their CRC along with the marine area, month, and day.

Ask to view **all** Catch Record Cards on this boat from the fishers that caught Dungeness crab and are responsible (or should be responsible) for recording the catch. Ask each question and record the following:

- k) **Total Number of Catch Record Cards (CRCs) Checked.** Select the total number of CRCs you checked. Ask for all CRCs from crabbers on this boat that retained Dungeness crab and are responsible for recording. Note: If you select 5+ CRCs checked, a new field will appear asking you to manually enter the number checked.
- l) **Total Number of Crab Recorded.** Enter the **total** number of crabs recorded across all CRCs from this boat. Use a handheld counter if it helps to tally the total number of recorded crabs. For an individual CRC, if there is a discrepancy between the number of crabs marked (X) and the number of crabs totaled, use the highest number between the two. Note: If no crabs are recorded or no fishers have their CRC in possession, enter 0.

Collecting Weight and Catch Compliance Data

Catch compliance = Checking for retention of female, soft-shell, and sub-legal crab.

The next set of questions will only appear if Dungeness and/or Red Rock crab are in possession on the vessel being interviewed. You should target approximately every **3rd** vessel for obtaining a weight sample and checking catch compliance of Dungeness crab retained. Do not collect weights or check for catch compliance (sex, legal size, soft-shell) on 'backed' crab, where the carapace (shell) has been removed. Sample all crabs from the intercepted vessel (i.e., do not sub-sample their catch) and treat Dungeness crab and Red Rock crab separately when collecting the information below.

Dungeness Crab:

- m) **Collecting a batch weight of Dungeness crab?** Select **Yes** or **No**.
- n) **Number of Dungeness Crab Weighed.** Enter the total number of Dungeness crab included in the batch weight.
- o) **Dungeness Crab Batch Weight:** Enter the total weight. Tally all weight measurements if multiple batch weights are taken. Ensure the bucket with the crab is completely drained of water before taking a weight. *Tare your scale, and/or make sure to subtract the weight of the bucket so it is not included in the recorded weight. Ensure you are in the correct units!!*
- p) **Checking Catch Compliance of Dungeness Crab?** Select **Yes** or **No**. Select 'Yes' if you are checking for female, sub-legal, or soft-shell.
- q) **Number of Dungeness Crab Checked for Compliance.** Enter the total number of Dungeness crab being checked for compliance.
- r) **Number of Female Dungeness Crab in Possession.** Select the total number of female Dungeness crab retained. Note: If you select 5+ female Dungeness retained, a new field will appear asking you to manually enter the number greater than 5.
- s) **Number of Sub-legal male Dungeness Crab in Possession.** Select the total number of sub-legal male Dungeness crab retained. Note: If you select 5+ sub-legal Dungeness

retained, a new field will appear asking you to manually enter the number greater than 5.

- t) **Number of soft-shell male Dungeness Crab in Possession.** Select the total number of soft-shell male Dungeness crab retained. Note: If you select 5+ soft-shell Dungeness retained, a new field will appear asking you to manually enter the number greater than 5.

Red Rock Crab:

- u) **Collecting a batch weight of Red Rock Crab?** Select Yes or No.
- v) **Number of Red Rock Crab Weighed.** Enter the total number of red rock crab included in the batch weight.
- w) **Red Rock Crab Batch Weight:** Enter the total weight. Tally all weight measurements if multiple batch weights are taken. Ensure the bucket with the crab is completely drained of water before taking a weight. *Tare your scale, and/or make sure to subtract the weight of the bucket so it is not included in the recorded weight. Ensure you are in the correct units!!*
- x) **Checking Catch Compliance of Red Rock Crab?** Select **Yes** or **No**. Select 'Yes' if you are checking for female or sub-legal.
- y) **Number of Red Rock Crab Checked for Compliance.** Enter the total number of red rock crab being checked for compliance.
- z) **Number of Female Red Rock Crab in Possession.** Select the total number of female red rock crab retained. Note: If you select 5+ female red rock retained, a new field will appear asking you to manually enter the number greater than 5.
- aa) **Number of Sub-legal Red Rock Crab in Possession.** Select the total number of sub-legal red rock crab retained. Note: If you select 5+ sub-legal red rock retained, a new field will appear asking you to manually enter the number greater than 5.
8. When you are finished sampling a boat, select **“Done”** at the bottom of the form. If you are done interviewing boats for the day, return to the “Header” form.
9. Once you arrive at the header form, complete the remaining header fields: **Creel End Time** and **Was enforcement present.** If you are ready to submit the form, select the **READY TO SUBMIT** toggle and hit **“Done”**.

Handing Out Education Materials

1. **Crab gauges.** Crab gauges are limited. Please only hand out gauges to crabbers that do not possess a gauge and/or are not properly measuring their crab.
2. **New crabber packets (brochure, rot cord, information cards):** New crabber packets are limited. Please give a crabber packet to harvesters that are new to the fishery or have indicated they don't fully understand certain rules or safe practices.

Appendix B

Appendix B. Sampling instructions for checking for soft-shell Dungeness crab.

1. Grab the crab from the back and turn it upside down so the abdomen of the crab is facing up.
2. Carefully push the 'elbow' of one claw towards the mouth of the crab, exposing the area of the shell that is typically covered by the claw.
3. Press the shell with your thumb at the point shown below, gradually increasing the pressure. If the shell flexes, bends, or gives in at approximately over 1 millimeter, it is a soft-shell crab. If the shell has not flexed and you have reached the point at which you would have easily crushed a tough peanut, then you have a hard (and legal) crab.
4. Soft-shell crab may also feel lighter than expected and be white on the abdomen. The shells of hard-shell crabs tend to be a darker yellowish brown and are often covered with barnacles and algae.

